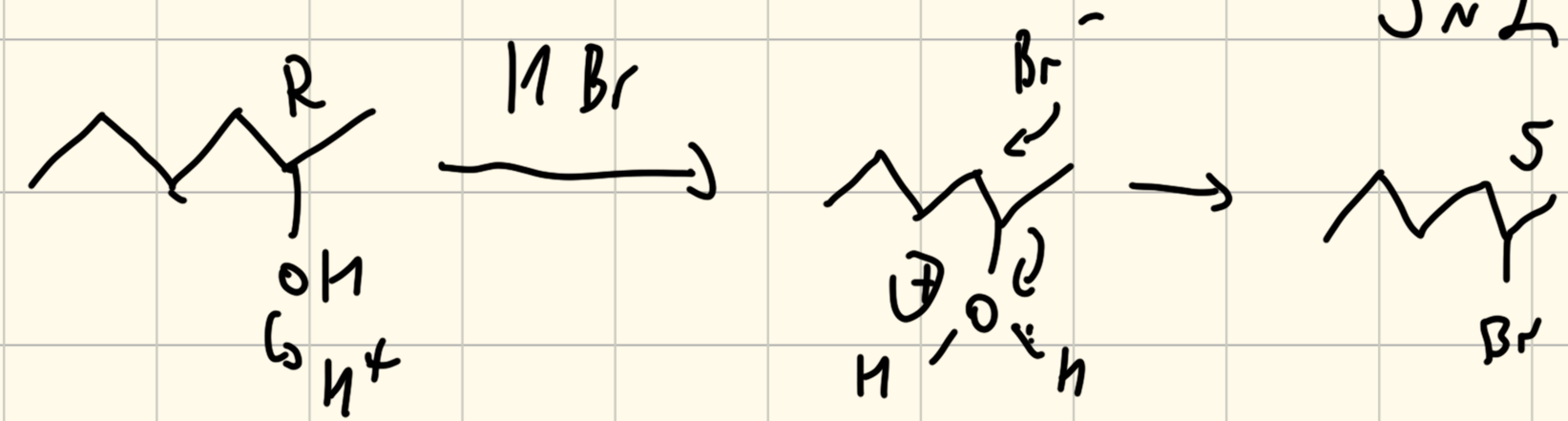
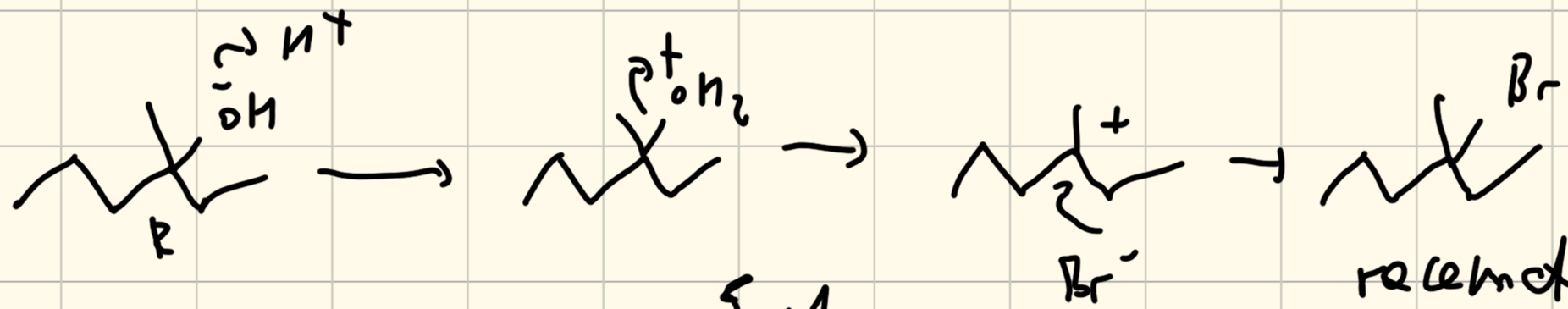
1)

a)



SN2

b)



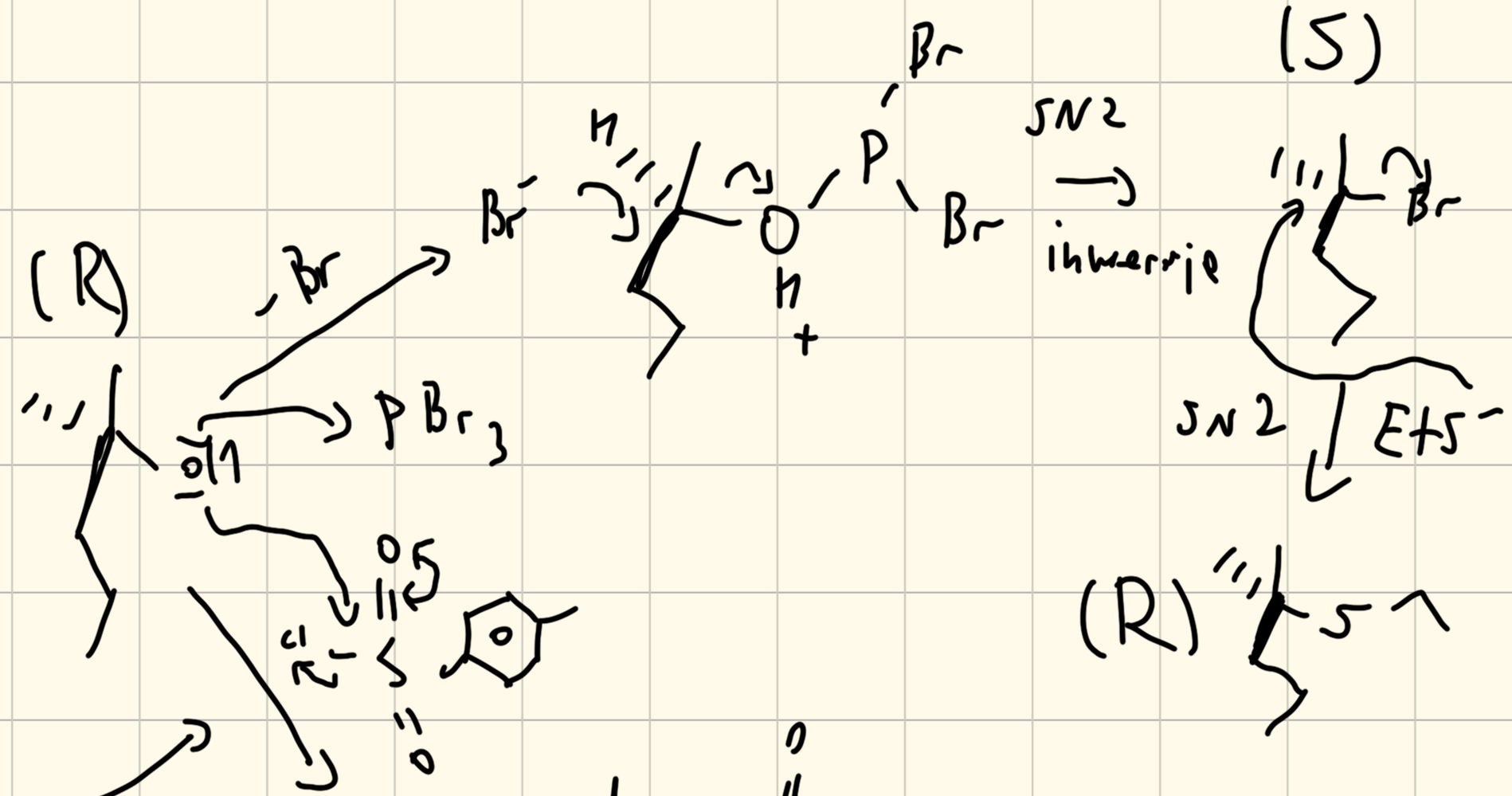
SN1

bo 3° alkohol

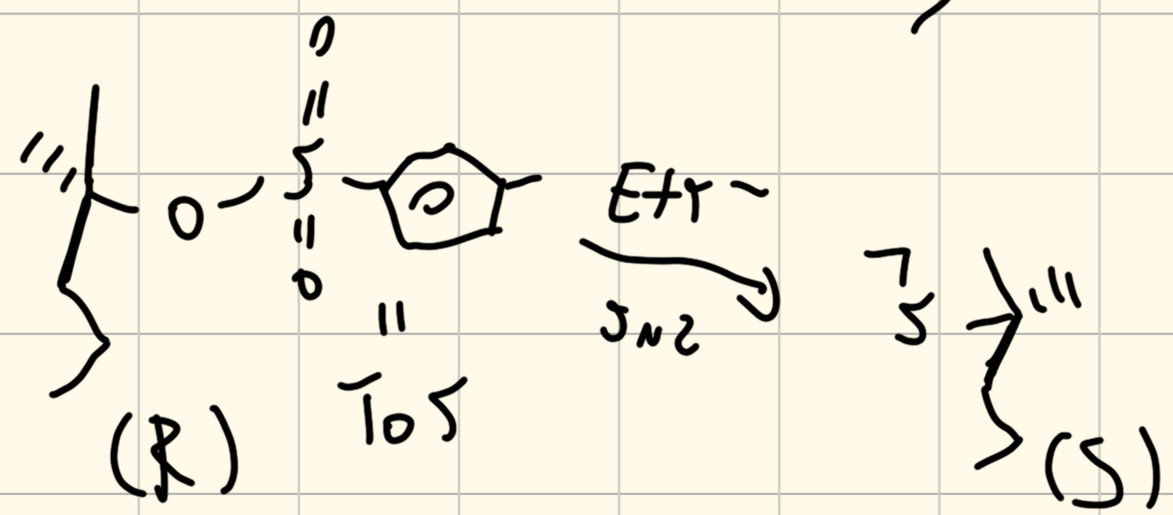
racemat

R : S = 50 : 50

2)



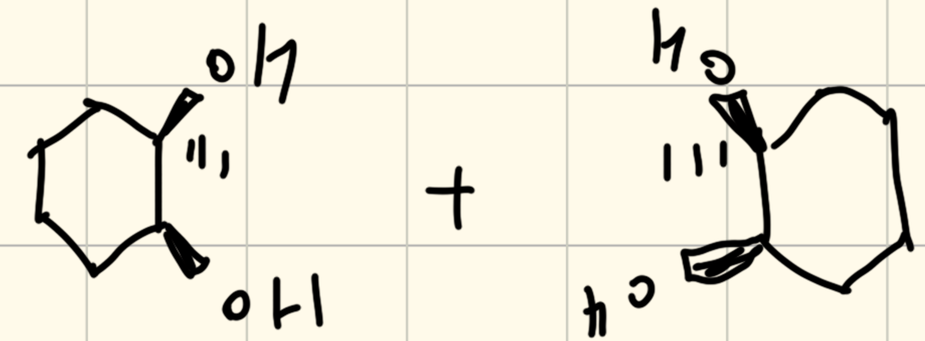
tutaj to alkohol jest nukleofilem, atakuje chlorek kwasowy



3)

1) OsO_4
2) $NaHSO_3$

50 : 50



syn

RCO_3H

50 : 50

+

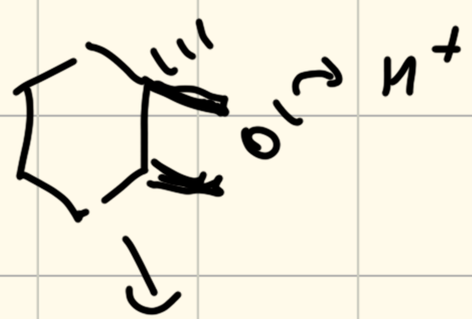
anti

(i)

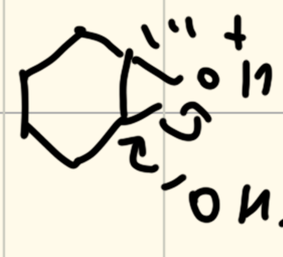
$SN2$

tu analogicznie

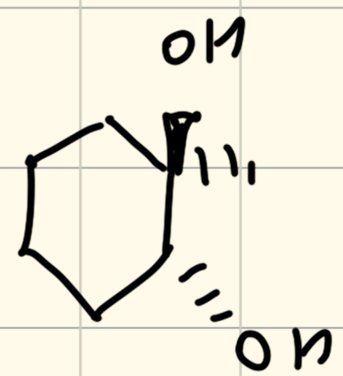
dokładny mechanizm (i):



H^+

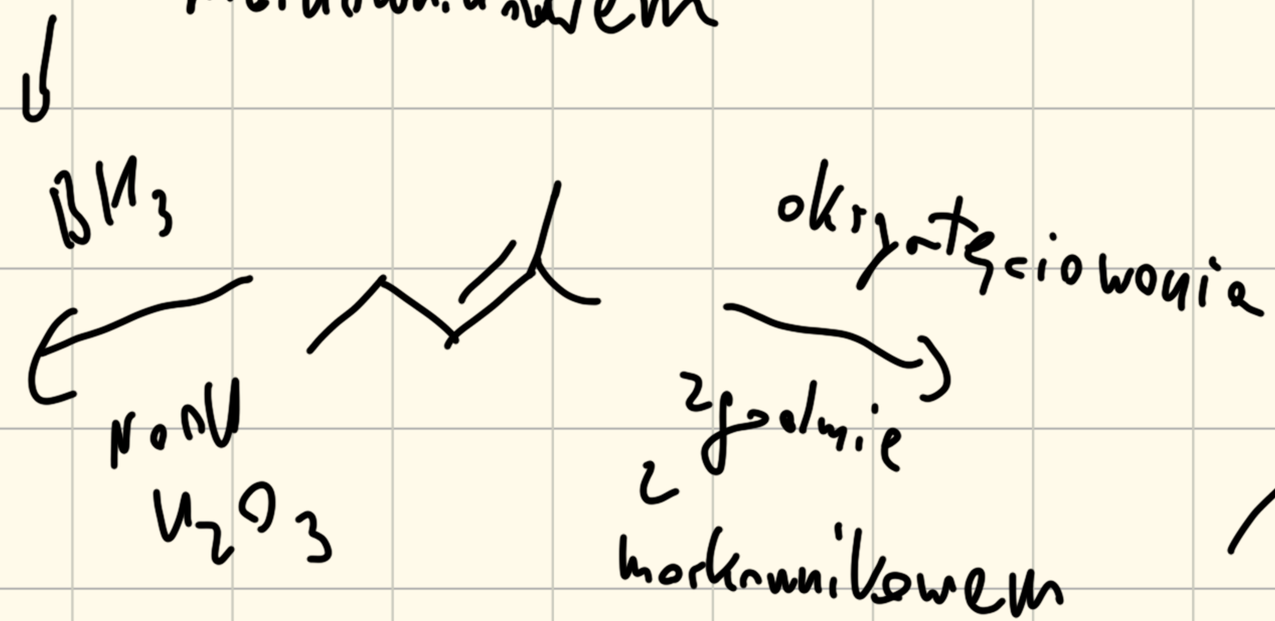


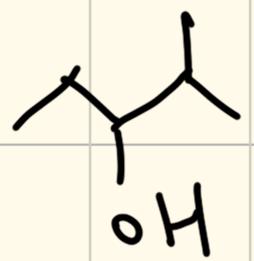
$-H^+$



4)

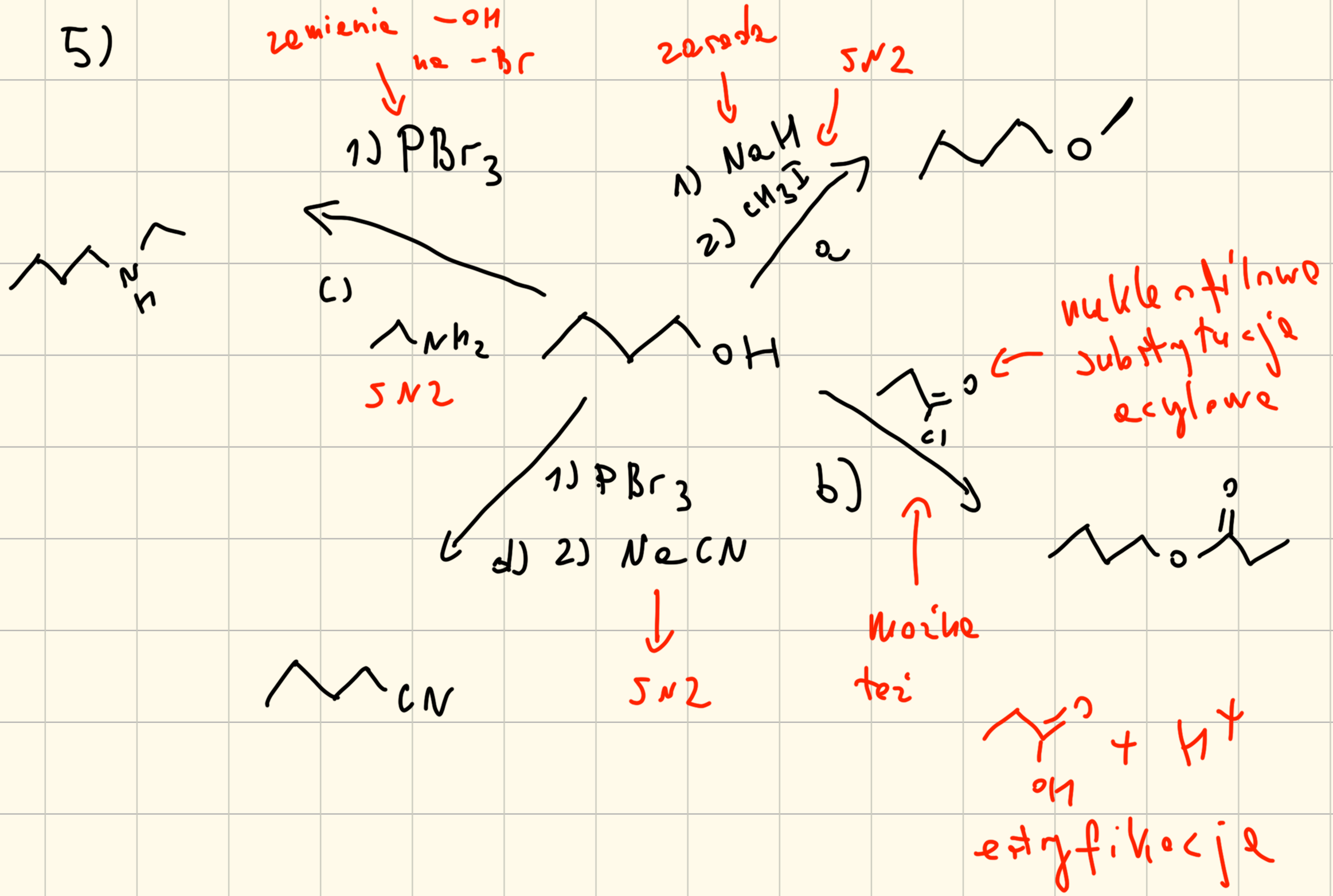
hydroborowanie niezgodnie z Markownikowem



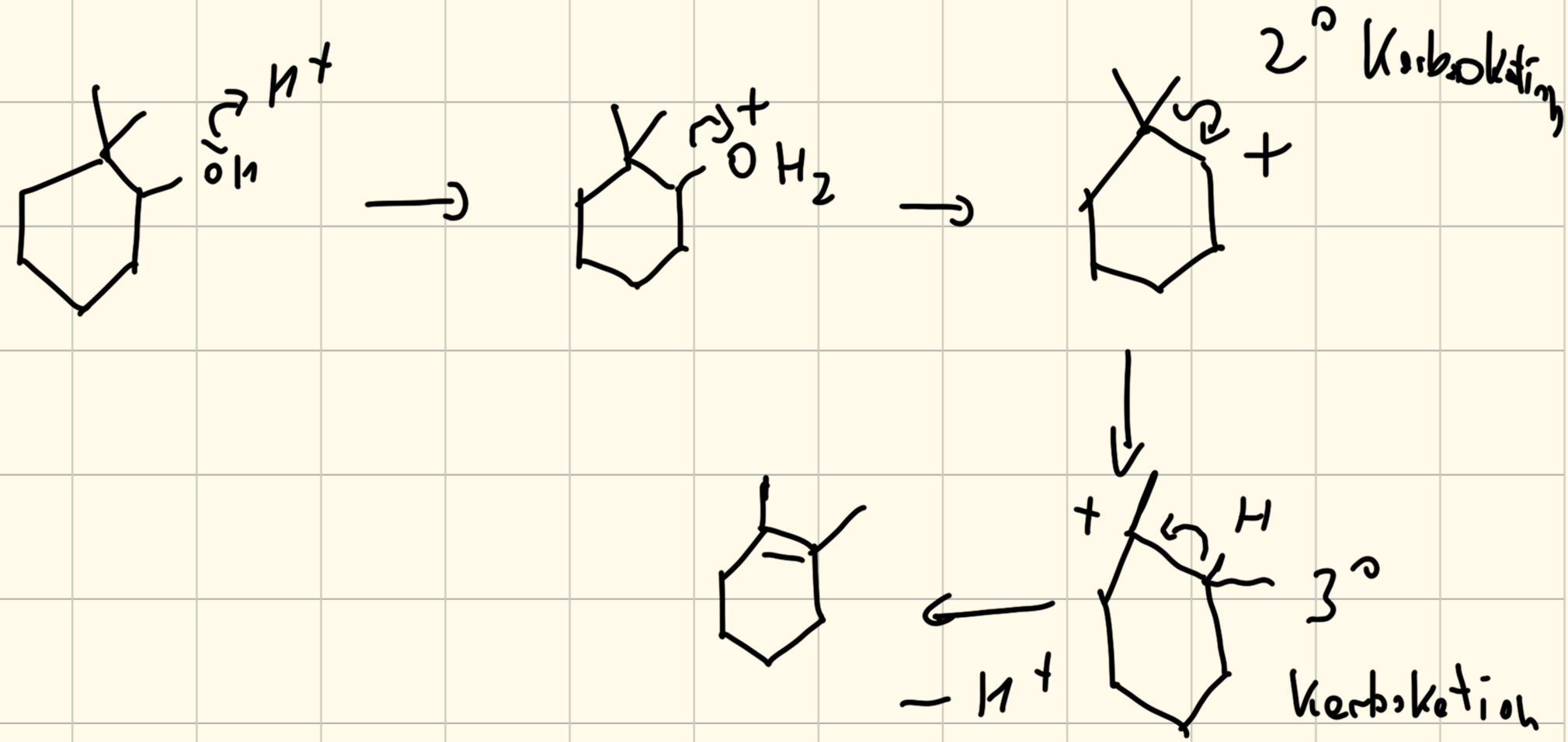


(racemat)

5)


zamienia -OH na -Br
1) PBr3
zasada
SN2
1) NaH
2) CH3I
a)
c)
NH2
SN2
OH
nukleofilowa substytucja acylowa
1) PBr3
d) 2) NaCN
b)
SN2
można też
CN
OH
+ H+
estryfikacja

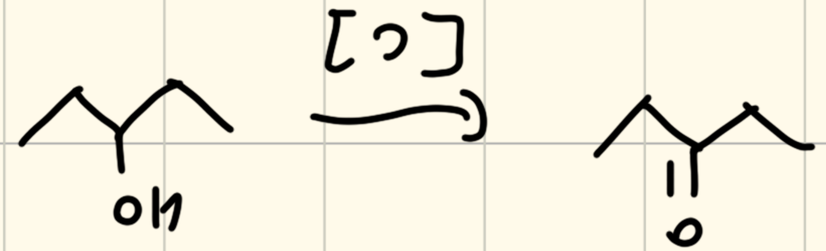
6) Przegrupowanie karbokationu



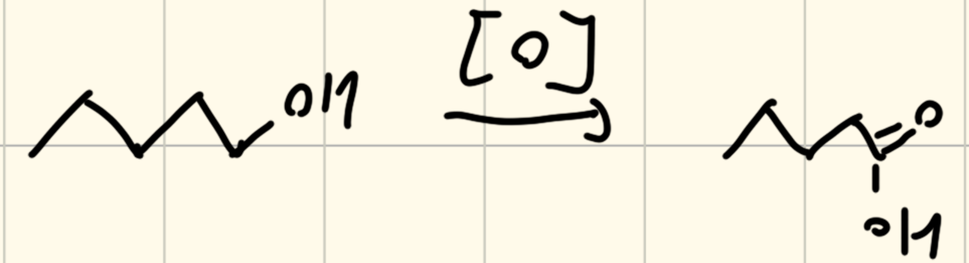
7)

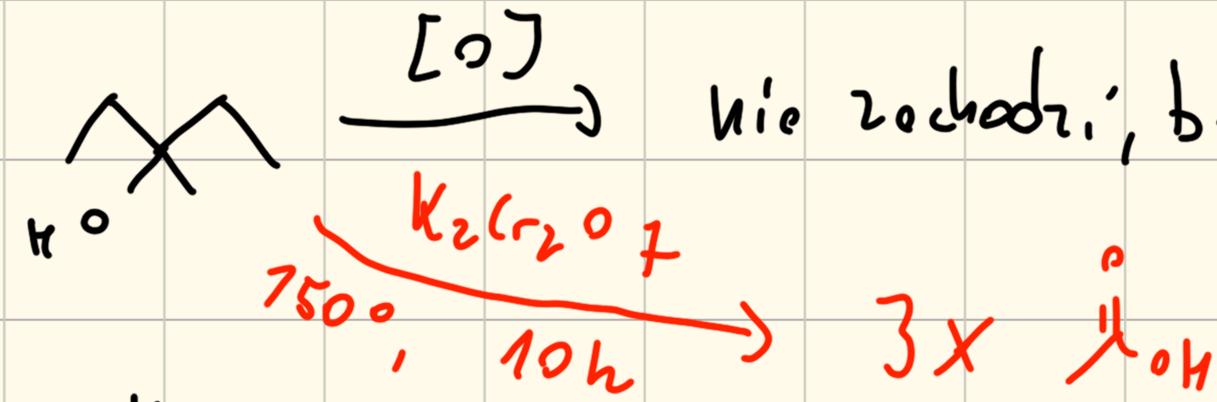
alkohol $\xrightarrow{1^\circ}$ aldehyd $\rightarrow$ kwas

$\xrightarrow{2^\circ}$ keton

a) 

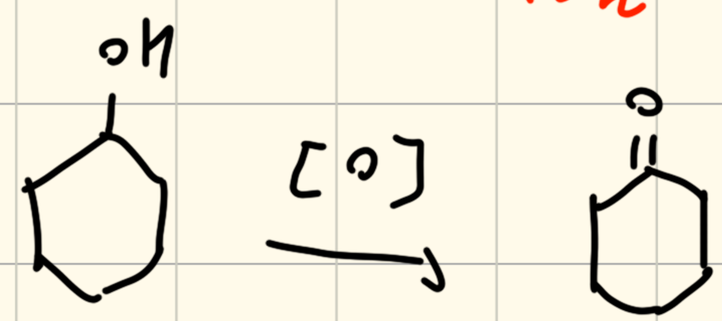
$[O] = K_2Cr_2O_7 + H^+$

b) 

3) 

nie zachodzi, bo 3° alko

surowe warunki rozcinają C-C

4) 

8)

$NaHCO_3$ – słabe zasada, na kwasy karboksylowe

$NaOH$ – silna zasada, na fenole

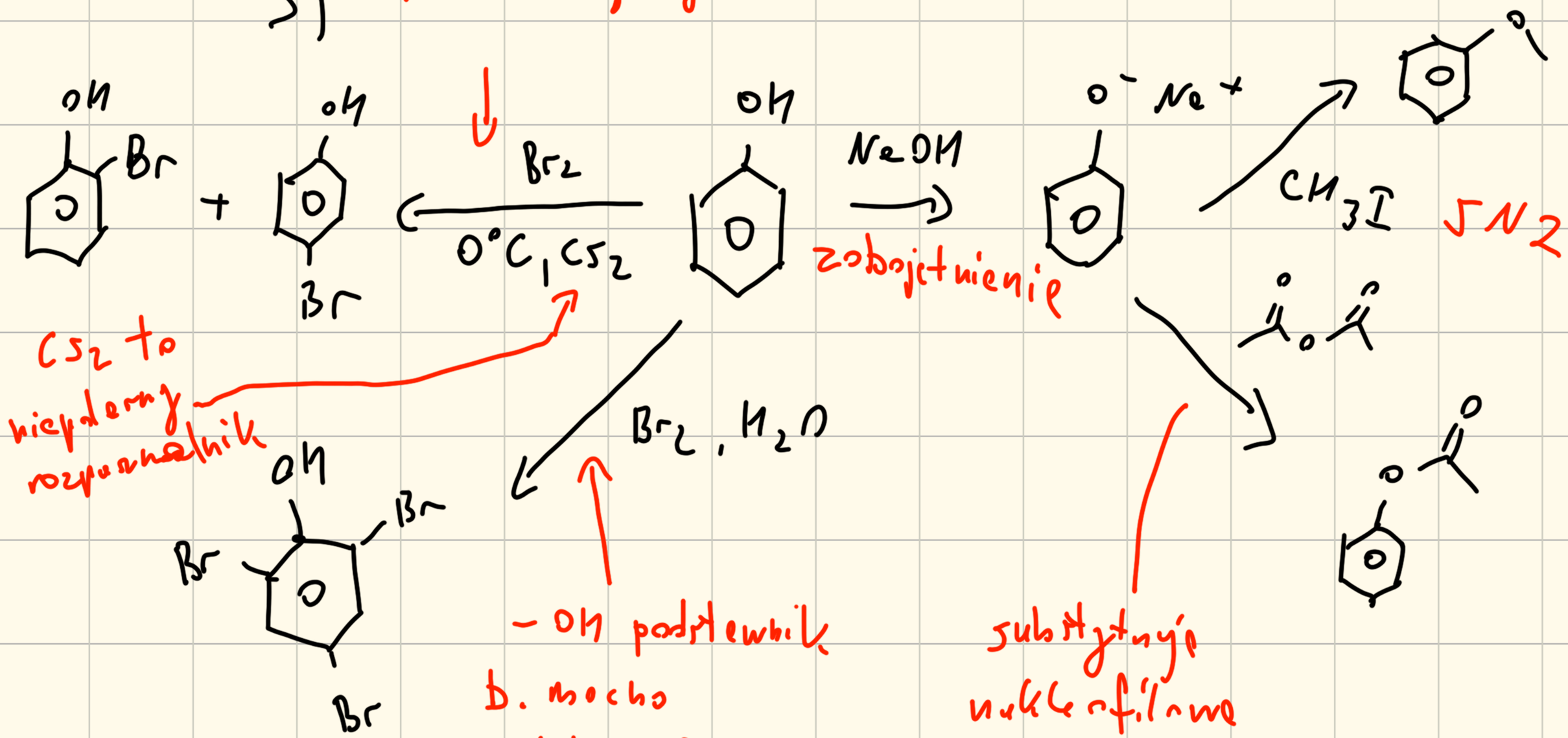
NaH – b. silna zasada, na alkohole

a) NaH

b) $NaOH$ (NaH też zadziała ale Overkill)

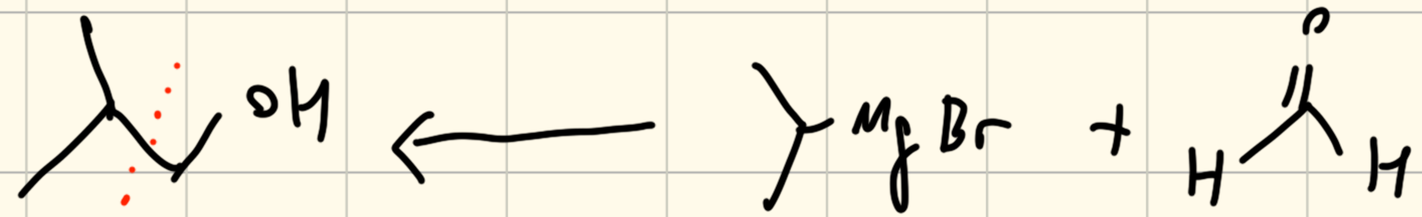
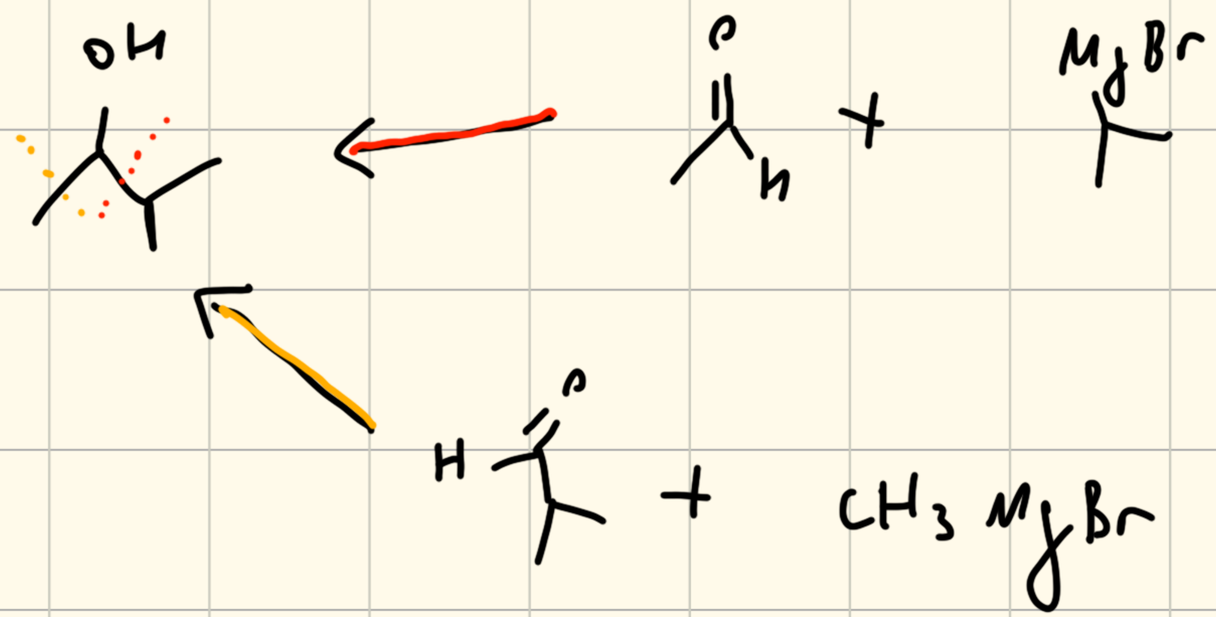
($NaHCO_3$ w b można, ale słabo reakcja zajdzie :P)

S_1 monosubstytuuje, bo 0°C



10)

a)



b)